Supplementary Information

Current Status and Future Developments in Computational Prediction of Synergistic Anticancer Drug Combinations

Yanhao Zhu[1,†], Di Su[2,3,†], Hua Wang[4,*], Juntao Liu[1,*]

[1]School of Mathematics and Statistics, Shandong University, Weihai, China

[2]Science and Technology Innovation Center, Shandong First Medical University and Shandong Academy of Medical Sciences, Jinan, Shandong, 250117, China

[3]Department of Respiratory Diseases, Shandong Second Provincial General Hospital (Shandong Province ENT Hospital), Shandong University, Jinan 250022, China

[4]Department of Neurology, Shandong Provincial Hospital Affiliated to Shandong First Medical University, Jinan 250021, China

[†]These authors contributed equally to this study

*To whom correspondence should be addressed

Email addresses:

HW: huawang@email.sdfmu.edu.cn

JL: juntaosdu@126.com

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Supplementary Note 1 | Calculation of drug combination synergy metrics

Drug responses can be reported using different phenotypic readouts, such as cell viability, growth, or inhibition, whose numerical scales and directions are not always consistent. To maintain a common notation, unless otherwise specified, the formulations below express drug response as fractional inhibition E ranging from 0 to 1, where $E = 0$ indicates no inhibition and $E = 1$ indicates complete inhibition. If fractional viability V is reported, it can be converted as $V = 1 - E$. Most classical drug-interaction metrics were originally formulated for pairwise combinations, and pairwise dose-response matrices remain the predominant experimental design. Therefore, the following equations are written for drugs A and B. Let d_A and d_B denote their respective doses, $E_A(d_A)$, $E_B(d_B)$, and $E_{AB}(d_A, d_B)$ denote the corresponding single-agent and observed combination effects. that explicitly define a non-interaction expectation E_{ref}, the pointwise synergy score is generally expressed as

$$S(d_A, d_B) = E_{AB}(d_A, d_B) - E_{\mathrm{ref}}(d_A, d_B)$$

A positive score indicates that the observed combination effect exceeds the corresponding non-interaction expectation, a score of zero indicates non-interaction under that reference model, and a negative score indicates antagonism. When a single score is required for an entire dose-response matrix, pointwise scores are commonly aggregated across the tested dose pairs, although the exact aggregation procedure may depend on the implementation.

Bliss independence. The Bliss independence model assumes that the two drugs exert their effects independently. At dose d_A and d_B, the fractions unaffected by drugs A and B are $1 - E_A(d_A)$ and $1 - E_B(d_B)$, respectively. Under independent action, the expected unaffected fraction of the combination is therefore their product, giving the expected combination inhibition

$$E_{\mathrm{Bliss}} = 1 - [1 - E_A(d_A)][1 - E_B(d_B)] = E_A(d_A) + E_B(d_B) - E_A(d_A)E_B(d_B).$$

The pointwise Bliss synergy score is

$$S_{\mathrm{Bliss}} = E_{AB}(d_A, d_B) - E_{\mathrm{Bliss}}.$$

Thus, $S_{\mathrm{Bliss}} > 0$ indicates synergy, $S_{\mathrm{Bliss}} = 0$ indicates Bliss independence, and $S_{\mathrm{Bliss}} < 0$ indicates antagonism. Because the expected effect is calculated directly from the two corresponding single-agent responses, no dose-response curve fitting is required.

Highest single agent. The highest single agent (HSA) metric uses the stronger effect of the two individual drugs at the corresponding doses as a benchmark for assessing whether the combination improves upon the better-performing monotherapy:

$$E_{\mathrm{HSA}} = \max[E_A(d_A), E_B(d_B)].$$

The pointwise HSA score is therefore

$$S_{\mathrm{HSA}} = E_{AB}(d_A, d_B) - E_{\mathrm{HSA}}.$$

A positive score indicates that the combination produces greater inhibition than either constituent drug alone, a score of zero indicates an effect equal to the better-performing monotherapy, and a negative score indicates that the combination is weaker than the better-performing monotherapy.

Loewe additivity. The Loewe additivity model is based on dose equivalence. Let $f_A(d)$ and $f_B(d)$ denote the fitted single-agent dose-response functions of drugs A and B, respectively. Their inverse functions,

$$D_A(E) = f_A^{-1}(E), \qquad D_B(E) = f_B^{-1}(E),$$

represent the doses of drugs A and B that would individually produce an inhibition level E. For the administered combination doses d_A and d_B, the expected Loewe-additive response E_{Loewe} is defined as the response satisfying

$$\frac{d_A}{D_A(E_{\mathrm{Loewe}})} + \frac{d_B}{D_B(E_{\mathrm{Loewe}})} = 1.$$

The two terms represent the fractions of the corresponding single-agent doses required to produce the same effect. Their sum being equal to one defines dose additivity. Because $D_A(E)$ and $D_B(E)$ are obtained from the single-agent dose-response curves, E_{Loewe} is generally determined numerically.

The pointwise Loewe synergy score is then

$$S_{\mathrm{Loewe}} = E_{AB}(d_A, d_B) - E_{\mathrm{Loewe}}.$$

Accordingly, $S_{\mathrm{Loewe}} > 0$ indicates synergy, $S_{\mathrm{Loewe}} = 0$ indicates Loewe additivity, and $S_{\mathrm{Loewe}} < 0$ indicates antagonism. The same dose-equivalence relationship can alternatively be evaluated at the observed combination effect rather than solved for the expected effect; this formulation forms the basis of the combination index described below.

Zero interaction potency. The zero interaction potency model defines non-interaction as

the absence of changes in the dose-response relationship of either drug when the other drug is present. ZIP therefore differs from the preceding pointwise models in that both the single-agent and combination dose-response relationships are fitted before the interaction score is calculated.

For each drug *i*, the single-agent dose-response curve can be represented by a four-parameter log-logistic function,

$$f_i(d) = E_{min,i} + \frac{E_{max,i} - E_{min,i}}{1 + (\frac{m_i}{d})^{\lambda_i}},$$

where d is the drug dose, $E_{min,i}$ and $E_{max,i}$ are the lower and upper response asymptotes, m_i is the dose corresponding to the midpoint between these asymptotes and represents the potency parameter, and λ_i determines the slope of the dose-response curve. The fitted single-agent responses at the tested doses are

$$\hat{E}_A = f_A(d_A), \qquad \hat{E}_B = f_B(d_B).$$

For normalized responses with common lower and upper limits, the expected response under zero interaction can be written as

$$E_{\mathrm{ZIP}} = \hat{E}_A + \hat{E}_B - \hat{E}_A\hat{E}_B.$$

This expression gives the expected combination response if the dose-response behavior of each drug remains unchanged in the presence of the other drug. However, the ZIP score is not obtained simply by subtracting this expectation from the raw observed combination response. Instead, the combination dose-response matrix is also fitted in two directions. At a fixed dose d_B of drug B, the responses across doses of drug A define a conditional dose-response curve for A, denoted $f_{A|B}$; similarly, at a fixed dose d_A of drug A, the responses across doses of drug B define $f_{B|A}$. These conditional fits yield the fitted combination responses

$$\hat{E}_{AB}^{A|B} = f_{A|B}(d_A; d_B), \qquad \hat{E}_{AB}^{B|A} = f_{B|A}(d_B; d_A).$$

The fitted conditional curves contain combination-specific midpoint and slope parameters, which can be compared with the corresponding single-agent parameters to capture changes in drug potency and dose-response shape. The pointwise ZIP delta score is the average deviation from the zero-interaction expectation obtained from the two fitting directions,

$$\delta(d_A, d_B) = \frac{1}{2}\left[\hat{E}_{AB}^{A|B} - E_{\mathrm{ZIP}}^{A|B} + \hat{E}_{AB}^{B|A} - E_{\mathrm{ZIP}}^{B|A}\right].$$

Under the ZIP reference model, the two directional expected responses are identical,

$$E_{\mathrm{ZIP}}^{A|B} = E_{\mathrm{ZIP}}^{B|A} = E_{\mathrm{ZIP}},$$

and the expression can therefore be written compactly as

$$\delta(d_A, d_B) = \frac{\hat{E}_{AB}^{A|B} + \hat{E}_{AB}^{B|A}}{2} - E_{\mathrm{ZIP}}.$$

A positive δ indicates synergy, $\delta = 0$ indicates zero interaction potency, and a negative δ indicates antagonism. For a dose-response matrix containing *N* tested combination-dose points, an overall ZIP score can be obtained by averaging the pointwise delta scores,

$$\Delta_{\mathrm{ZIP}} = \frac{1}{N}\sum_{p,q} \delta\left(d_{A,p}, d_{B,q}\right).$$

The compact formulation above describes the calculation logic without expanding all fitted parameters. The complete pairwise delta expression in terms of the single-agent and combination-specific midpoint and slope parameters m_A, m_B, $m_{A|B}$, $m_{B|A}$, λ_A, λ_B, $\lambda_{A|B}$ and $\lambda_{B|A}$ is given in Equation 19 of Yadav et al[1]. The corresponding derivation for the full four-parameter log-logistic model with explicit lower and upper response asymptotes is provided in Appendix A3 of the same study.

NCI-ALMANAC ComboScore. The original NCI-ALMANAC ComboScore is defined on the NCI percentage-growth scale and therefore retains its original response definition rather than being converted to the fractional-inhibition notation above. Let G_A^p and G_B^q denote the percentage growth observed after treatment with drug A at concentration *p* and drug B at concentration *q*, respectively, and let G_{AB}^{pq} denote the observed percentage growth under their combination. On this scale, 100 represents growth comparable with the untreated control, 0 represents no net growth relative to the time-zero measurement, and negative values indicate net cell loss. First, define the truncated single-agent growth values as

$$\tilde{G}_A^p = \min\left(G_A^p, 100\right), \qquad \tilde{G}_B^q = \min\left(G_B^q, 100\right).$$

The expected combination growth is

$$G_{\mathrm{exp}}^{pq} = \begin{cases} \min\left(G_A^p, G_B^q\right), & G_A^p \le 0 \; or \; G_B^q \le 0, \\ \dfrac{\tilde{G}_A^p \tilde{G}_B^q}{100}, & \text{otherwise.} \end{cases}$$

The ComboScore is then calculated by summing the difference between expected and observed growth over all tested concentration pairs,

$$\text{ComboScore} = \sum_{p,q} (G_{\text{exp}}^{pq} - G_{AB}^{pq}).$$

Because lower growth represents a stronger treatment effect, a positive contribution indicates that the observed combination suppresses cell growth more strongly than expected. Thus, a positive ComboScore indicates greater-than-additive activity, zero indicates the expected activity, and a negative score indicates less-than-additive activity. This definition follows the original NCI-ALMANAC formulation.

Combination index. The combination index evaluates the same general dose-equivalence relationship used in Loewe additivity from a different direction. Suppose doses d_A and d_B administered in combination produce an observed inhibition E_{AB}. Let $D_A(E_{AB})$ and $D_B(E_{AB})$ denote the doses of drugs A and B that would individually be required to produce the same inhibition E_{AB}. The combination index is

$$CI = \frac{d_A}{D_A(E_{AB})} + \frac{d_B}{D_B(E_{AB})}.$$

Each term therefore represents the fraction of the corresponding single-agent equivalent dose used in the combination. CI < 1 indicates that the observed effect is achieved using less than the additive equivalent dose and therefore indicates synergy; CI = 1 indicates additivity; and CI > 1 indicates antagonism.

Supplementary Note 2 | Evaluation metrics for computational drug combination prediction

Let y_i and $\hat{y}_i$ denote the experimentally measured and predicted values, respectively, for sample *i*, and let *n* denote the total number of samples. For classification tasks, the positive class is assumed to represent synergistic combinations unless otherwise specified.

Metrics for regression tasks

Root mean squared error (RMSE) measures the average magnitude of prediction errors while assigning greater weight to large deviations:

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}.$$

A smaller RMSE indicates closer agreement between predicted and experimentally measured

values.

Mean absolute error (MAE) measures the average absolute difference between predicted and observed values:

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|.$$

Compared with RMSE, MAE is less strongly influenced by individual large errors.

Coefficient of determination, R^2, quantifies the proportion of variation in the observed values explained by the predictions:

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2},$$

where

$$\bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i.$$

An R^2 value of 1 indicates perfect agreement. A value of 0 indicates prediction performance equivalent to using the mean observed value as a constant prediction, whereas negative values indicate poorer performance than this baseline.

Pearson correlation coefficient measures the linear association between predicted and observed values:

$$r = \frac{\sum_{i=1}^{n}(y_i - \bar{y})(\hat{y}_i - \bar{\hat{y}})}{\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}\sqrt{\sum_{i=1}^{n}(\hat{y}_i - \bar{\hat{y}})^2}},$$

where $\bar{y}$ and $\bar{\hat{y}}$ are the means of the observed and predicted values, respectively. Pearson *r* ranges from -1 to 1, with values closer to 1 indicating stronger positive linear agreement.

Spearman rank correlation coefficient evaluates whether predicted and observed values have a consistent monotonic rank ordering. It is calculated as the Pearson correlation between their ranks:

$$\rho = \mathrm{corr}\big(R(y_i), R(\hat{y}_i)\big).$$

where corr() represents the calculation of the Pearson correlation coefficient between the two. $R(y_i)$ and $R(\hat{y}_i)$ denote the ranks assigned to the observed and predicted values, respectively, after ordering each set of values in the same ascending or descending direction. When there are no tied ranks, the coefficient can also be written as

$$\rho = 1 - \frac{6\sum_{i=1}^{n} d_i^2}{n(n^2-1)},$$

where $d_i = R(y_i) - R(\hat{y}_i)$ is the difference between the observed and predicted ranks of sample *i*. A value close to 1 indicates that samples ranked high experimentally also tend to be ranked high by the model, regardless of whether the predicted numerical values exactly match the observed scores.

Metrics for classification tasks

For binary classification, let true positives (TP) denote synergistic combinations correctly predicted as synergistic, false positives (FP) denote nonsynergistic combinations incorrectly predicted as synergistic, true negatives (TN) denote nonsynergistic combinations correctly predicted as nonsynergistic, and false negatives (FN) denote synergistic combinations incorrectly predicted as nonsynergistic.

Accuracy (ACC) measures the proportion of correctly classified samples:

$$\mathrm{ACC} = \frac{TP + TN}{TP + TN + FP + FN}.$$

Although intuitive, accuracy can be dominated by the majority class when synergistic and nonsynergistic samples are strongly imbalanced.

Precision measures the proportion of predicted synergistic combinations that are truly synergistic:

$$\mathrm{Precision} = \frac{TP}{TP + FP}.$$

High precision indicates a low false-positive rate among combinations selected as synergistic.

Recall, also termed sensitivity or true-positive rate, measures the proportion of experimentally synergistic combinations that are successfully identified:

$$\mathrm{Recall} = \frac{TP}{TP + FN}.$$

High recall indicates that few true synergistic combinations are missed.

F1 score is the harmonic mean of precision and recall:

$$F_1 = 2\frac{\mathrm{Precission} \times \mathrm{Recall}}{\mathrm{Precission} + \mathrm{Recall}}.$$

Equivalently,

$$F_1 = \frac{2TP}{2TP + FP + FN}.$$

It provides a single measure that balances false-positive and false-negative errors without

directly incorporating the number of true negatives.

Area under the receiver operating characteristic curve (AUROC). The receiver operating characteristic (ROC) curve evaluates classification performance across different decision thresholds. For each threshold, the true-positive rate and false-positive rate are calculated as

$$\text{TPR} = \frac{TP}{TP + FN},$$
$$\text{FPR} = \frac{FP}{FP + TN}.$$

The classification threshold is varied across the range of prediction scores, producing a series of (FPR, TPR) points that form the ROC curve. The area under this curve, AUROC, is then calculated numerically from these points, commonly using trapezoidal interpolation between adjacent points. AUROC ranges from 0 to 1. A value of 1 indicates perfect discrimination, whereas 0.5 corresponds to random ranking. AUROC can also be interpreted as the probability that a randomly selected synergistic combination receives a higher prediction score than a randomly selected nonsynergistic combination.

Area under the precision-recall curve (AUPRC). The precision–recall (PR) curve is constructed in a similar manner by varying the classification threshold and calculating precision and recall at each threshold:

$$\text{Precision} = \frac{TP}{TP + FP},$$
$$\text{Recall} = \frac{TP}{TP + FN}.$$

The resulting (Recall, Precision) points form the PR curve, and the area under this curve is summarized as AUPRC. In practice, the area is calculated numerically from the sequence of precision–recall points generated as the threshold changes. AUPRC emphasizes performance on the positive class and is therefore particularly informative when synergistic combinations are much less frequent than nonsynergistic combinations. Unlike AUROC, whose random baseline is 0.5, the expected AUPRC under random prediction depends on the proportion of positive samples in the dataset.

Interpretation across data-splitting settings

The same numerical metric can represent substantially different prediction problems

depending on how the data are partitioned. Under random splitting, the same drugs and cell lines may occur in both training and test sets, so performance primarily reflects interpolation within an observed entity space. Holding out entire drug pairs, individual drugs, or cell lines instead evaluates generalization to unseen combinations, drugs, or cancer contexts, respectively. Cross-dataset evaluation further measures transfer between independently generated screening datasets. Metrics should therefore always be interpreted together with the corresponding data-splitting and generalization setting.

Supplementary Tables

Supplementary Table S1 | Computational models based on direct phenotypic mapping

Model	Prediction task	Combination data	Drug information	Cancer-context information	Additional biological/pharmacological information	Modeling strategy/method	Interpretability
DeepSynergy[2]	Synergy-score regression	O'Neil	Molecular fingerprints; chemical descriptors	Gene expression	—	Independently encodes the two drugs and cell line, concatenates their representations, and predicts synergy with a feed-forward DNN	No explicit model interpretation
Narayan et al.[3]	Synergy classification	Literature-curated combinations; AstraZeneca-Sanger/DREAM	Single-agent response profiles	Driver mutations	Cross-cell-line single-agent response profiles and drug-target annotations	Builds a drug atlas from cross-cell-line single-agent responses and predicts synergy using logistic regression based on the distance between the two drugs on the atlas, their sensitivities in the given cell line, and drug target–mutation matches.	Interpretable features based on drug distance, sensitivity, and target–mutation matching
AuDNNsynergy[4]	Synergy-score regression	O'Neil	Molecular fingerprints; chemical descriptors	Gene expression;Gene mutations; copy-number alterations	—	Pretrains autoencoders on gene-expression, mutation, and copy-number data, then combines the encoded cancer context with drug features in a DNN for synergy prediction.	No explicit model interpretation
MatchMaker[5]	Synergy-score regression	DrugComb	Chemical descriptors	Gene expression	—	Conditions each drug on the cell-line expression profile in two shared drug-specific subnetworks, then combines the two context-specific drug representations to predict synergy.	No explicit model interpretation
DeepDDS[6]	Synergy classification	O'Neil; AstraZeneca-Sanger/DREAM	Molecular graphs	Gene expression	—	Encodes each drug molecular graph with GAT/GCN and cell-line expression with fully connected layers, then fuses the representations for synergy classification.	GAT-based atom and substructure importance analysis
PRODeepSyn[7]	Synergy-score regression	O'Neil	Morgan fingerprint; chemical descriptors	Gene expression; Gene mutations	PPI network	Uses a PPI-based GCN to integrate cell-line gene expression and mutations into cell embeddings, then combines them with drug fingerprints and descriptors in a DNN.	Cell-line embedding visualization and distance analysis
EC-DFR[8]	Synergy-score regression	DrugComb	Molecular fingerprints; chemical descriptors	Gene expression	Drug-induced gene-expression profiles	Uses an enhanced cascade deep forest that dynamically resamples hard-to-predict samples across layers and combines layer outputs by weighted voting.	Feature contribution, SHAP analysis, and KEGG/GO enrichment of key genes
DTSyn[9]	Synergy classification	O'Neil; AstraZeneca-Sanger/DREAM; FLOBAK; NCI-ALMANAC; FORCINA; YOHE	Molecular graphs	Gene expression	PPI-derived gene embeddings	Uses a fine-grained Transformer for drug-substructure–gene and gene–gene interactions and a coarse-grained Transformer for drug–drug and drug–cell interactions before synergy prediction.	Attention-based analysis of drug–cell and drug-substructure–gene interactions
SYNPRED[10]	Synergy-score regression + synergy classification	NCI-ALMANAC	Chemical descriptors	Gene expression; copy-number alterations; DNA methylation; chromatin profiling; metabolomics; miRNA; proteomics	—	Stacks multiple machine-learning and DNN base learners to predict drug-combination effects under several synergy metrics.	Permutation importance with biological annotation of top omics features
SNRMPACDC[11]	Synergy-score regression	O'Neil	Molecular fingerprints; chemical descriptors	Gene expression; mutations; copy-number alterations	—	Encodes the two drugs with a shared Siamese CNN, forms a symmetric drug-pair representation through random projection and element-wise multiplication, and combines it with CNN-encoded cell features for prediction	No explicit model interpretation

DEML[12]	Synergy-score regression + synergy classification + adverse DDI classification	DrugComb	Chemical descriptors	Gene expression	Adverse drug–drug interaction labels	Uses a multi-gate mixture-of-experts framework to jointly learn multiple synergy tasks and adverse drug–drug interactions.	Deep SHAP attribution with key-gene and biological-process analysis
HyperSynergy[13]	Synergy-score regression	SYNERGxDB	Molecular graphs	Gene expression	—	Uses a Bayesian prior-guided hypernetwork to generate cell-line-specific synergy predictors, enabling zero- and few-shot adaptation to data-poor cell lines	No explicit model interpretation
DFFNDDS[14]	Synergy classification	DrugComb; DrugCombDB	SMILES; molecular fingerprints	Gene expression	—	Encodes SMILES with a pretrained language model and combines them with atom-pair fingerprints and cell-line features through attention-based feature fusion for synergy prediction.	No explicit model interpretation
Zhang et al.[15]	Synergy-score regression	NCI-ALMANAC; O'Neil; FORCINA; Mathews	Drug identity; molecular fingerprints	Gene expression	Single-agent dose-response profiles	Harmonizes single-agent dose–response curves across studies by interpolation and imputation, then combines these features with drug and cell information in LightGBM to predict combination response and synergy.	SHAP analysis of feature and dose-response contributions
MARSY[16]	Synergy-score regression + single-agent response prediction	DrugComb	Drug perturbation signatures	Gene expression	Single-agent drug responses	Learns separate drug-pair and drug-pair–cell-line representations and jointly predicts synergy together with the responses of both single drugs in a multitask network.	Latent embedding visualization by PCA; no feature-level attribution
EXPRESS[17]	Synergy-score regression	Beat AML	Drug identity	Gene expression (patient)	Drug–target annotations	Trains an ensemble of XGBoost models on patient molecular profiles and averages both predictions and Shapley attributions to obtain stable synergy predictions and explanations.	Ensemble SHAP, Shapley interaction analysis, and pathway/signature enrichment
CGMS[18]	Synergy-score regression + single-agent response prediction	O'Neil	Molecular fingerprints; chemical descriptors	Gene expression	Single-agent sensitivity	Represents the two drugs and cell line as a three-node heterogeneous complete graph, learns an order-invariant whole-graph representation with hierarchical attention, and jointly predicts synergy and monotherapy sensitivity.	Attention-weight and attention-masking analysis
RECOVER[19]	Synergy-score regression	NCI-ALMANAC	Drug identity; molecular fingerprints	Fixed cell line (MCF7)	—	Uses a shared drug encoder and symmetric interaction predictor with deep-ensemble uncertainty in a sequential optimization loop that iteratively selects new combinations for testing.	Latent-space analysis of drug embeddings against known targets and mechanisms of action
MTLSynergy[20]	Synergy-score regression + single-agent response prediction	O'Neil	Molecular fingerprints; chemical descriptors	Gene expression	Single-agent sensitivity	Pretrains drug and cell representations and uses shared features with task-specific heads to jointly predict combination synergy and monotherapy sensitivity.	No explicit model interpretation
SynergyX[21]	Synergy-score regression	DrugComb	SMILES-derived substructures	Gene expression; mutations; copy-number alterations; DNA methylation; gene effect; gene dependency	PPI and drug-target information for biologically informed gene selection	Encodes multi-omics with convolution-augmented attention and uses bidirectional mutual attention to model drug-substructure–cell interactions before synergy prediction.	Drug–cell attention maps, pharmacophore/substructure highlighting, DTI overlap, and pathway enrichment
CancerGPT[22]	Synergy classification	DrugComb	Drug identity	Cell-line/tissue identity	Single-agent sensitivity	Converts drug-pair and cellular-context data into natural-language prompts and fine-tunes a GPT-2-based model for zero- and few-shot synergy classification in rare tissues.	No CancerGPT-specific explanation; separate GPT-3-generated rationales were checked against literature
MFSynDCP[23]	Synergy classification	O'Neil; AstraZeneca-Sanger/DREAM	Molecular graphs	Gene expression	—	Uses GATs to encode drug graphs, adaptive attention pooling to emphasize interaction-relevant substructures, and a collaborative drug–cell feature-interaction module for synergy classification.	Attention-based visualization of key chemical substructures

PDSP[24]	Synergy-score regression + single-agent response prediction	DrugComb; He et al. patient ex vivo dataset	Chemical descriptors	Gene expression (cell line/patient)	Single-agent responses	Jointly learns drug-pair synergy and single-drug response in cell lines, then fine-tunes the model with patient gene expression and limited ex vivo single-drug measurements for patient-specific prediction.	No explicit model interpretation
MultiComb[25]	Synergy-score regression + combination-sensitivity prediction	O'Neil	Molecular graphs	Gene expression	Combination sensitivity scores	Encodes drug graphs with GCNs and cell-line expression with fully connected networks, then uses attention and a cross-stitch multitask module to jointly predict synergy and combination sensitivity.	No explicit biological interpretation; attention is used for task-specific feature weighting
JointSyn[26]	Synergy-score regression	O'Neil; NCI-ALMANAC	Molecular graphs; molecular fingerprints	Gene expression	—	Learns two complementary views from a joint drug-pair graph and fingerprint-based drug-cell representations, then fuses them for synergy prediction and few-shot adaptation.	No explicit feature attribution; pan-cancer case analyses relate predictions to tissue type and EGFR mutation status
Pourmousa et al.[27]	Synergy-score regression + synergy classification	PANC-1 screen	Molecular fingerprints; chemical descriptors; molecular graphs	Fixed cell line (PANC-1)	Single-agent responses; mechanism-of-action annotations	Uses three independent machine learning pipelines trained on PANC-1 screening data to rank large numbers of virtual drug pairs and prioritize top candidates for experimental testing.	Mechanism-of-action analysis of predicted and experimentally validated combinations
GraphTranSynergy[28]	Synergy classification	O'Neil; AstraZeneca-Sanger/DREAM	Molecular graphs	Gene expression	—	Encodes drug molecular graphs with a residual Graph Transformer and cell-line gene expression with BiLSTM, then fuses the learned representations for synergy classification.	No explicit model interpretation
FRSynergy[29]	Synergy classification	DrugCombDB; Oncology-Screen	Molecular graphs	Gene expression	Drug-protein interactions; cell-protein interactions; PPI network; side effects	Learns topology-based drug and cell-line representations with a heterogeneous graph attention network, then refines these features according to each drug-drug-cell-line context before synergy prediction.	No explicit biological interpretation; feature refinement remains data-driven
BAITSAO[30]	Synergy-score regression + synergy classification + single-agent response prediction	DrugComb; O'Neil	Drug text embeddings	Cell-line text embeddings	Single-agent inhibition	Uses LLM-derived drug and cell-line representations and pretrains a shared network on selected synergy and single-drug-response tasks with uncertainty-weighted multitask learning, followed by zero-shot or fine-tuned transfer.	SHAP gene attribution with differential-expression and pathway enrichment analyses
Pisces[31]	Synergy classification	GDSC-combo; DrugCombDB	SMILES; molecular graphs; 3D molecular embeddings; drug text embeddings	Gene expression	Drug targets, side effects, single-agent drug responses, and drug ontology	Projects multiple drug modalities into a shared space, expands each drug pair into many modality-pair views, and aggregates these views to predict synergy while accommodating missing modalities.	Genetic-interaction-network analysis identifies biologically relevant drug-sensitive pathways
MultiSyn[32]	Synergy classification	O'Neil; AstraZeneca-Sanger/DREAM	Molecular graphs	Gene expression; mutations	PPI network	Integrates PPI and multi-omics data for cell representations, learns drug features from atom-fragment heterogeneous graphs with a graph Transformer, and fuses them for synergy classification.	Fragment/atom attention visualization of key pharmacophore substructures and representation analysis
SynProtX[33]	Synergy-score regression + synergy classification	DrugComb	Molecular graphs; molecular fingerprints	Gene expression; proteomics	—	Combines GAT-based molecular-graph features and molecular fingerprints with cell-line transcriptomic and proteomic profiles for synergy prediction.	Integrated Gradients for protein/gene attribution and GAT attention-based chemical-substructure analysis
BridgeSyn[34]	Synergy-score regression	O'Neil; NCI-ALMANAC	3D molecular embeddings	Gene expression	Protein-sequence embeddings from ProtBert	Encodes drugs with pretrained molecular models and enriches cell representations with protein-language-model information, then uses shared bridge tokens and cross-attention to fuse the drug-cell-drug triplet.	Perturbation-based drug contribution analysis and Integrated Gradients attribution
MetaSynergy[35]	Synergy-score regression	SYNERGxDB	Molecular graphs	Gene expression	—	Combines drug molecular-graph and cell-line omics encoders with cross-domain pretraining and meta-optimization, followed by few-shot fine-tuning in data-scarce cell lines	No explicit model interpretation
CoSynLLM[36]	Synergy-score regression	NCI-ALMANAC; O'Neil	Drug text embeddings; molecular fingerprints	Gene expression	—	Combines LLM-generated chemical descriptions with molecular fingerprints and cell-line gene expression, then progressively integrates the three feature types through a hierarchical fusion network.	No explicit model-level attribution; LLM-generated descriptions provide human-readable chemical context.

UniSyn[37]	Synergy-score regression + single-agent response prediction	DrugComb; O'Neil; NCI-ALMANAC	Molecular fingerprints; SMILES embeddings; molecular graphs	Gene expression;gene mutations; copy-number alterations	Single-agent sensitivity	Fuses multimodal drug and cell-line features with self- and cross-attention and transfers knowledge from monotherapy sensitivity to synergy through multitask knowledge transfer.	Self-/cross-attention analysis, mechanism-of-action case studies, and pathway enrichment of synergy-sensitive cell clusters
DALI-Syn[38]	Synergy classification	O'Neil; AstraZeneca-Sanger/DREAM	SMILES	Gene expression	—	Encodes SMILES with ChemBERTa-2 and multi-scale convolution, refines drug and cell features with self-attention, and uses bidirectional cross-attention to model drug-cell interactions before prediction.	Cross-attention visualization of cell-specific drug-gene interaction patterns
DeepDrugs[39]	Synergy-score regression	DrugComb; O'Neil	Molecular graphs	Gene expression;gene mutations; copy-number alterations	—	Encodes drug molecular graphs and cell-line multi-omics and uses tri-linear attention to jointly model drug A–drug B–cell interactions before contextual fusion and synergy regression.	Tri-linear attention identifies cell-specific pharmacophores; high-attention atoms are quantitatively compared with annotated binding sites
CADS[40]	Synergy-score regression	DrugComb	SMILES; molecular graphs	Gene expression;gene mutations; copy-number alterations; gene dependencies	—	Adds learnable causal and trivial gene masks to existing synergy models and uses backdoor adjustment with parallel supervision to separate putatively causal cell-line signals from confounding features.	Gene-level causal scores, causal-gene masking, Integrated Gradients, representation analysis, and KEGG/GSEA pathway interpretation
DGBDF[41]	Synergy classification	Jaaks et al.	Molecular fingerprints	Gene expression	—	Uses a cascade deep forest on drug fingerprints and cell-line expression with hardness- and uncertainty-guided undersampling, allowing successive layers to focus on difficult and ambiguous samples under severe class imbalance.	SHAP analysis of drug fingerprint bits, cell-line genes, and fingerprint-gene interactions
MetaComb[42]	Synergy classification	DrugComb; He et al. patient ex vivo dataset	Molecular graphs; molecular fingerprints	Gene expression (cell line/patient)	—	Encodes drugs with GCNs plus Morgan fingerprints and cell/patient expression with an MLP, then applies MAML across cell-line-specific tasks to learn an initialization that can be few-shot fine-tuned to data-poor cell lines or individual patients.	No explicit model interpretation

Note. An em-dash indicates that no information is used.

Supplementary Table S2 | Computational models based on relational inference

Model	Prediction task	Combination data	Drug information	Cancer-context information	Additional biological/pharmacological information	Modeling strategy/method	Interpretability
DTF[43]	Synergy classification	O'Neil	Drug identity	Cell-line identity	—	Organizes observed drug-drug-cell-line synergy into a tensor, factorizes it with CP-WOPT to learn latent drug and cell representations, and uses these representations in a DNN to predict unmeasured combinations.	No explicit model interpretation
TuSDC[44]	Combination link prediction	Literature-derived disease-drug-drug dataset	Drug identity	Cancer identity	Disease comorbidities and drug–disease treatment relations extracted from biomedical literature	Integrates known drug combinations, disease comorbidities, and drug–disease treatments extracted from biomedical literature, and jointly factorizes these relations with Tucker tensor decomposition to learn drug and disease representations and rank drug pairs for a given disease.	No explicit model interpretation
Hypergraph Synergy[45]	Synergy-score regression + synergy classification	O'Neil; NCI-ALMANAC	Molecular graphs	Gene expression	—	Represents drugs and cell lines as nodes and known synergistic drug-drug-cell-line triplets as hyperedges, then uses hypergraph propagation together with drug/cell features and similarity reconstruction to predict unmeasured triplets.	Hypergraph topology and node-degree analysis identify influential drugs/cell lines and relational patterns
GAECDS[46]	Synergy classification	DrugComb	Molecular fingerprints	Gene expression	—	Learns drug representations from the known drug-synergy graph using a graph autoencoder and combines them with cell-line features in a CNN to predict unobserved drug–cell-line combinations.	No explicit model interpretation
MGAE-DC[47]	Synergy-score regression + synergy classification	O'Neil; NCI-ALMANAC; CLOUD; FORCINA	Molecular fingerprints	Gene expression	—	Constructs separate cell-line-specific drug graphs for synergistic, additive, and antagonistic relationships and uses multi-channel graph autoencoders to learn relation-aware drug representations for synergy prediction.	Visualization and analysis of cell-line-specific drug embeddings and relational patterns
Hypertran Synergy[48]	Synergy-score regression + synergy classification	O'Neil; NCI-ALMANAC	Molecular graphs	Gene expression	—	Builds a drug-drug-cell-line hypergraph and uses a Transformer to learn coarse-grained multi-way relations, while similarity reconstruction captures fine-grained drug and cell information before feature fusion and synergy prediction.	Coarse- versus fine-grained representation analysis; no direct molecular attribution
MHCLSyn[49]	Synergy classification	O'Neil; NCI-ALMANAC	Molecular graphs	Gene expression	—	Constructs a drug-synergy hypergraph and generates drug-replacement, cell-line-replacement, and random-perturbation views, then uses hypergraph contrastive learning to obtain robust representations for synergy prediction.	No explicit biological interpretation
HyperSynergyX[50]	Synergy classification	DCDB 2.0; C-DCDB; DrugMAP 2.0	Molecular fingerprints; Anatomical Therapeutic Chemical classification codes	Cancer identity	—	Represents known three-drug combinations as hyperedges, uses a dual-biased random walk to capture relationships among drug triplets, and applies tensor decomposition to score unobserved three-drug combinations.	KG-RAG retrieves relevant drug–target and pathway relations and generates biologically grounded mechanistic hypotheses for predicted combinations

Note. An em dash indicates that no information is used.

Supplementary Table S3 | Computational models based on target- and network-mediated modeling

Model	Prediction task	Combination data	Drug information	Cancer-context information	Additional biological/pharmacological information	Modeling strategy/method	Interpretability
Cheng et al.[51]	Combination prioritization	Curated clinical and literature combinations	Drug targets	Disease-associated proteins	Drug targets, disease-associated proteins, and human PPI network	Maps drug targets and disease proteins onto the human interactome and prioritizes combinations whose targets jointly cover the disease module while occupying separated network neighborhoods.	Drug-target proximity, disease-module coverage, and network separation provide direct topological interpretation
DrugCell[52]	Single-agent response regression + combination prioritization	GDSC; CTRP; DeepSynergy; PDX Encyclopedia	Molecular fingerprints	Gene mutations	Gene Ontology hierarchy of cellular subsystems	Uses a visible neural network organized by Gene Ontology-defined cellular subsystems to model genotype-dependent drug response and identify complementary functional vulnerabilities for combination design	RzLIPP identifies GO-defined cellular subsystems that drive drug response and suggests complementary targetable processes
TranSynergy[53]	Synergy-score regression	O'Neil	Drug targets	Gene expression; gene dependencies	Drug-target interactions and PPI network	Propagates known drug targets through a PPI network to generate gene-level drug-action profiles and integrates them with cell-line expression and dependency in a Transformer for synergy prediction	Shapley-based gene attribution and pathway enrichment identify genes and pathways underlying predicted synergy
ComboNet[54]	Synergy classification + single-agent response prediction + DTI prediction	NCATS; Riva et al.; Bobrowski et al.; HIV combination dataset	Molecular graphs	Fixed viral-infection context	Drug-target interactions and single-agent antiviral activity	Jointly models drug–target interactions and target–disease associations using drug structures, biological networks, single-drug antiviral activity, and limited combination data to predict synergistic antiviral pairs.	Predicted drug–target interactions provide target-level mechanistic interpretation
GraphSynergy[55]	Synergy classification	DrugCombDB; Oncology-Screen	Drug targets	Cell-line-associated proteins	Drug-protein and cell-line-protein associations plus PPI network	Places drug targets and cell-associated proteins in a PPI network and combines learned graph representations with drug-cell module proximity and drug-drug target-module relationships to predict synergy.	Protein contribution analysis and PPI-based drug-cell/drug-drug module relationships
Muthene[56]	Synergy-score regression + adverse DDI classification	DrugComb; TWOSIDES	Molecular fingerprints	Gene expression	Drug/protein interaction network and adverse drug-drug effects	Learns drug representations from heterogeneous drug-target and adverse-effect networks, using biologically defined meta-paths and multitask learning to jointly predict synergy and adverse effects.	Meta-path analysis traces predictions through drug-target–protein relationships and associated biological processes
KGANSynergy[57]	Synergy classification	DrugCombDB; Oncology-Screen	Drug identity	Cell-line identity	Drug-protein, cell-line-protein, cell-line-tissue, and PPI networks	Builds a knowledge graph linking drugs, proteins, cell lines, and tissues and uses hierarchical neighborhood propagation with multi-head attention to learn representations for synergy prediction.	Knowledge-graph neighborhoods and attention weights indicate the relative contributions of neighboring biological entities
DGSSynADR[58]	Synergy-score regression	DrugComb	Molecular fingerprints;Drug identity	Gene expression	Drug-drug interactions, drug-target interactions, and PPI network	Builds a drug-target-protein heterogeneous graph, learns drug and target representations with message passing and global attention, and applies a cell-line-conditioned bilinear predictor.	No explicit model interpretation
LGSyn[59]	Synergy-score regression + synergy classification	O'Neil; Oncology-Screen	Molecular fingerprints; chemical descriptors	Gene expression	Biological relations encoded in a drug-protein-cell-line knowledge graph	Combines local drug fingerprints/descriptors and cell-line expression with global drug/cell representations learned by attention-based propagation over a biological knowledge graph, then fuses both levels in a DNN.	No explicit model interpretation
DeepTraSynergy[60]	Synergy classification	DrugCombDB; Oncology-Screen; O'Neil	SMILES	Cell–protein association profiles	Drug-target interactions, PPI network, and cell-target interactions	Uses Transformer-based drug representations together with drug-target, PPI, and cell-target information in a multitask network that predicts synergy while jointly learning drug-target interaction and toxicity as auxiliary tasks.	No explicit model interpretation

Jiang et al.[61]	Combination prioritization	Literature-derived cancer dataset	Drug targets	Cancer identity	Drug targets and cancer-specific PPI networks	Constructs cancer-specific protein interactomes and identifies combinations using drug-target network proximity and correlation clustering of functional communities.	Cancer-specific network communities, key genes, and pathway analysis provide mechanistic context
PathSynergy[62]	Synergy classification	O'Neil; DrugCombDB	Molecular graphs; drug targets	Gene expression	Drug–target interactions and signaling pathways	Maps drug targets and cell-line molecular features onto signaling pathways, learns pathway-aware representations with graph neural networks, and predicts synergy from drug-pathway-cell interactions.	Gradient-based pathway contribution analysis identifies pathways driving individual predictions
CASynergy[63]	Synergy classification	DrugCombDB; Oncology-Screen	Molecular graphs	Gene expression	PPI network and cell-line-protein associations for constructing cell-line-specific gene networks	Constructs cell-line-specific PPI networks and combines drug-cell cross-attention with causal feature masking to separate prediction-relevant molecular signals from shortcut features.	Causal attention and soft masks identify prediction-associated genes, subgraphs, and biological pathways
Xiao et al.[64]	Synergy classification	DrugCombDB; TTD; DrugBank; DDInter	Drug targets; drug-target regulatory effects	Disease-associated targets	Drug-target interactions and directed inter-target regulatory interactions	Constructs a directed weighted drug-target regulatory network that propagates activating and inhibitory drug effects to downstream targets, then predicts combination effects from relative distances between regulated target sets.	Signed regulatory paths and positive/negative target sets directly trace the network basis of predicted synergy or antagonism
SYNERGIE[65]	Combination prioritization	DrugCombDB	Drug targets; drug perturbation signatures	Patient transcriptomics	Drug–disease and drug-drug interactions plus disease-specific therapeutic pathways/targets	First learns general drug representations from a large biological knowledge graph, then adapts them on drug-combination-centric subgraphs to predict disease indications for known or unseen drug combinations.	Drug-combination-centric subgraphs retain explicit drug-target-pathway-disease relationships
DualKG-DC[66]	Drug-combination indication prediction	Literature-derived combination–disease dataset	Drug identity	Cancer identity	Drug targets, PPI network, pathways, gene-disease/phenotype relations, and disease comorbidities	Pretrains a biomedical knowledge graph, constructs combination-centered local subgraphs, and uses heterogeneous GNNs and link prediction to infer drug-combination indications	No explicit model interpretation

Supplementary Table S4 | Computational models based on dose-dependent combination-response modeling

Model	Prediction task	Combination data	Drug information	Cancer-context information	Additional biological/pharmacological information	Modeling strategy/method	Interpretability
comboFM[67]	Combination-response regression	NCI-ALMANAC	Drug identity; molecular fingerprints	Cell-line identity; gene expression	Drug doses	Uses the two drugs, their doses, and the cell line in a higher-order factorization machine to predict individual combination responses, which are assembled into complete dose–response matrices for synergy calculation.	No explicit model interpretation
comboLTR[68]	Combination-response regression	NCI-ALMANAC	Drug identity; molecular fingerprints	Cell-line identity; gene expression; copy-number alterations; gene dependencies; proteomics	Drug doses	Formulates drug-dose-drug-dose-cell response prediction as polynomial regression and uses latent tensor reconstruction to predict complete dose-response matrices, including previously untested combinations.	No explicit model interpretation
PIICM[69]	Combination-response regression	O'Neil	Drug identity	Cell-line identity	Drug doses and experimental response uncertainty	Uses a permutation-invariant Gaussian-process model to learn continuous combination dose-response surfaces while preserving symmetry between the two drugs and quantifying predictive uncertainty.	Gaussian-process posterior provides dose-specific predictive uncertainty; the response surface shows dose-dependent interaction patterns
SynBa[70]	Combination-response regression	DREAM; NCI-ALMANAC	—	—	Drug doses; optional prior knowledge through Bayesian parameter priors	Fits a two-dimensional generalized Hill model to combination dose–response data and uses Bayesian inference to estimate posterior distributions of response and synergy parameters.	Pharmacologically interpretable parameters for synergistic potency and efficacy together with posterior uncertainty
BATCHIE[71]	Combination-response regression	NCI-ALMANAC; GDSC2; MERCK; sarcoma screen	Drug identity	Cell-line identity	Drug doses and sequentially acquired combination-response measurements	Iteratively selects the most informative drug–dose–cell experiments using Bayesian active learning, updates a hierarchical Bayesian tensor-factorization model after each batch, and uses its posterior uncertainty to design the next experiments.	Explicitly decomposes combination response into single-drug effects and an interaction term; no molecular interpretation
DD-PRiSM[72]	Combination-response regression	NCI-ALMANAC; O'Neil	Molecular fingerprints	Gene expression	Single-agent dose-response profiles	Pretrains dose-dependent monotherapy response models and predicts combination response as weighted contributions from the two monotherapies plus a learned synergy component	Explicitly decomposes combination efficacy into each monotherapy contribution and the additional synergistic interaction across doses
DeepSynBa[73]	Combination-response regression	NCI-ALMANAC; O'Neil	SMILES embeddings	Gene expression	Drug doses	Predicts pharmacologically meaningful parameters of a two-dimensional dose-response model from drug and cell features and uses them to reconstruct the complete combination response surface.	Intermediate parameters separately quantify synergistic potency, maximal efficacy, and synergistic efficacy
Jeong et al.[74]	Combination-response regression	Wagoner et al. SARS-CoV-2 dataset	Drug identity	Fixed viral-infection context	Single-drug dose-response parameters and viral-dynamics parameters governing infectivity and viral production	Fits sparse diagonal combination measurements with Bliss pharmacodynamic and mechanistic viral-dynamics models, then reconstructs the full dose–response surface and quantifies synergy relative to Bliss independence	Viral-dynamics parameters link drug effects to viral infectivity and production, while an interaction parameter quantifies synergy

Note. An em dash indicates that no information is used.

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